Journal of Long-Term Effects of Medical Implants, 29(2):125–133 (2019)

Influences of Stresses on Dental Implant Threads Using Finite-Element Analysis

Vinamra Jain,[a] Vamsi Krishna Dommeti,[a] Masud Rana,[b] & Sandipan Roy[a,*]

[a]Department of Mechanical Engineering, SRM Institute of Science and Technology, Kattankulathur, Chennai 603203, India; [b]Department of Aerospace Engineering and Applied Mechanics, Indian Institute of Engineering Science and Technology, West Bengal-711103, India

*Address all correspondence to: Sandipan Roy, Department of Mechanical Engineering, SRM Institute of Science and Technology, Kattankulathur, Chennai 603203, India; Tel: +91 8584966296; Fax: +91 04427453903, E-mail: sandipan888roy@gmail.com

ABSTRACT: The objective of this finite-element analysis (FEA) study is to check the effect of stress on different types of threaded implants. To perform the FEA study of 81 models, the geometry of cancellous bone is imported and corresponding materials are set using FEA software. A common biting force is applied on the implant. After applying the load, each model is compared on the basis of equivalent von Mises stress. From the analysis, we observe that a trapezoidal type of thread profile is most suitable for dental implants.

KEY WORDS: dental implant, threads, dental biomechanics, FEA

I. INTRODUCTION

Numerous dental inserts and materials are used to make reclamations of projections and crowns; thus, dentists have many options to realize the best treatment for the particular needs of each of their patients. The most commonly used dental inserts are endosteal bones that are surgically embedded into jaw bone, providing an elective for bridges or separable dentures. The dental implant system has benefited from titanium's biocompatibility and osteoinductive properties that have led to proper osteointegration and improvement of the implant system.[1] This osteointegration phenomenon results in a high level of success for current treatments.

At the osseointegration level, a good outcome depends on an assortment of variables. A dental implant that is improperly placed influences the steady anchorage of its embedding into bone tissue. Some researchers have underscored the role of surface treatment for long-term solidity of inserts.[2] This appears to be closely related to the satisfactory rebuilding of implants, with bone restoration having an imperative role in the success of the embedded system.[3] Effective implant treatment generally depends on good primary stability (a mechanical phenomenon) and secondary stability (a biological phenomenon called osseointegration),[4] with victory of the embedded treatment depending by and large on solidity and auxiliary steadiness. Variables that play a major part in accomplishing the essential solidity of dental implants have been detailed in the literature.[4] Researchers have measured the solidity of dental inserts using reverberation recurrence examination[5] and *in vivo* mandibular implant-retained overdentures.[6] Larger surface area of bone–implant contact encourages stress loosening due to stack exchange along the bone–implant interface. Studies on behavior of short implants were performed to assess the relationships among implant factors such as length, diameter, thread pitch, and shape.[7,8] Additionally, bone-mending reactions to diverse embedded root shape plans as well as aggregate survival rates of brief inserts have been explored.[9–11] Progressed plans of level root shapes[12–14] have expanded total survival rate to > 90% and have led to advancements in organic reactions of early endosseous periembedded recuperation.[15,16] For resultant stress at the bone–implant interface, the main determinant is the design of an implant complimenting proper loading condition for better long-term implant stability. Excessive stress generation at the interface may cause bone resorption, resulting in better long-term implant stability.[17–19] Several research groups have worked on implant design using computational techniques, but it has not been validated that a bet-

1050-6934/19/$35.00 © 2019 by Begell House, Inc. www.begellhouse.com

ter osseointegration process occurs as a result of an optimum three-dimensional (3D) structure.[20,21]

The objective of the present investigation is to check mechanical stresses on the dental implant by varying parameters, including implant diameter, implant length, and implant thread type.

II. MATERIALS AND METHODS

To perform this finite-element analysis (FEA) study, the geometry of cancellous bone was imported into the computer, and properties of corresponding materials were set using ANSYS Workbench 18.1 (Canonsburg, PA). Solidworks 2018 (Dassault Systèmes, Vélizy-Villacoublay, France) was used for analyses of FEA models (27 combinations were used, as shown in Table 1, with three different materials, namely, stainless steel, Ti6Al4V, and commercially pure titanium). Different combinations of implant diameters, lengths, and thread shapes formed the 81 (27 combinations, each with three materials; i.e., 27 × 3 = 81) models.

The 3D model of cancellous bone was scanned and imported to Solidworks 2018, and a dental implant was designed. The inner bone is cancellous,

TABLE 1: Thread specifications

Type of thread	Length (mm)	Diameter (mm)
1 × 3–mm Rectangular	9	3
1 × 3–mm Rectangular	10	3
1 × 3–mm Rectangular	11	3
1 × 3–mm Rectangular	9	3.5
1 × 3–mm Rectangular	10	3.5
1 × 3–mm Rectangular	11	3.5
1 × 3–mm Rectangular	9	4
1 × 3–mm Rectangular	10	4
1 × 3–mm Rectangular	11	4
1 × 3–mm Rectangular	9	3
1 × 3–mm Rectangular	10	3
1 × 3–mm Rectangular	11	3
1 × 3–mm Rectangular	9	3.5
1 × 3–mm Rectangular	10	3.5
1 × 3–mm Rectangular	11	3.5
1 × 3–mm Rectangular	9	4
1 × 3–mm Rectangular	10	4
1 × 3–mm Rectangular	11	4
60º Trapezoid	9	3
60º Trapezoid	10	3
60º Trapezoid	11	3
60º Trapezoid	9	3.5
60º Trapezoid	10	3.5
60º Trapezoid	11	3.5
60º Trapezoid	9	4
60º Trapezoid	10	4
60º Trapezoid	11	4

and the outer bone with a 2-mm thickness is cortical. The assembly was imported in ANSYS Workbench 18.1 as a .STEP file. The material properties of cortical and cancellous bone were then assigned.[22] We made the assumption in this study that all materials were linear, elastic, homogeneous, and isotropic.[23–26]

The 81 different implant models ranged in length from 9 to 11 mm and in diameter from 3 to 4 mm. The rectangular threads had 1- and 2-mm widths, and the trapezoidal thread was at an angle of 60º, as shown in Figs. 1–3. The cross section of dental implant assembly can be seen in Fig. 4. Boundary conditions of the assembly side faces were fixed[22] (Fig. 5), and a compression load of 150 N as shown in Fig. 6 was applied to the top middle node of every assembly.

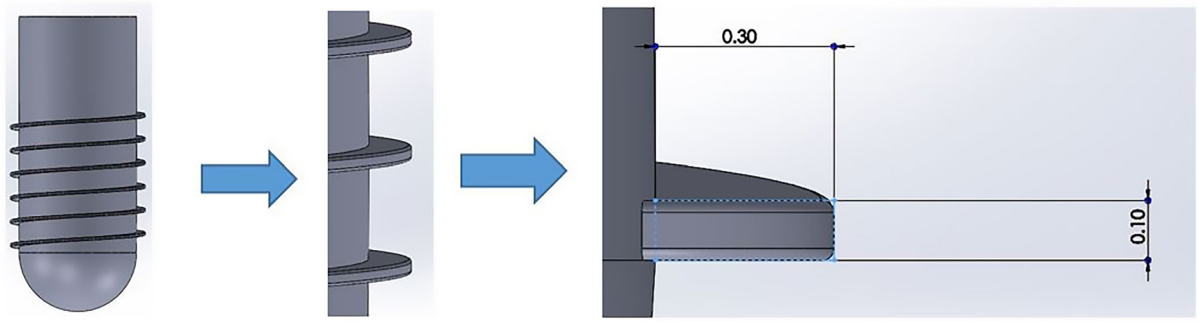


FIG. 1: 1 × 3–mm Implant thread profile

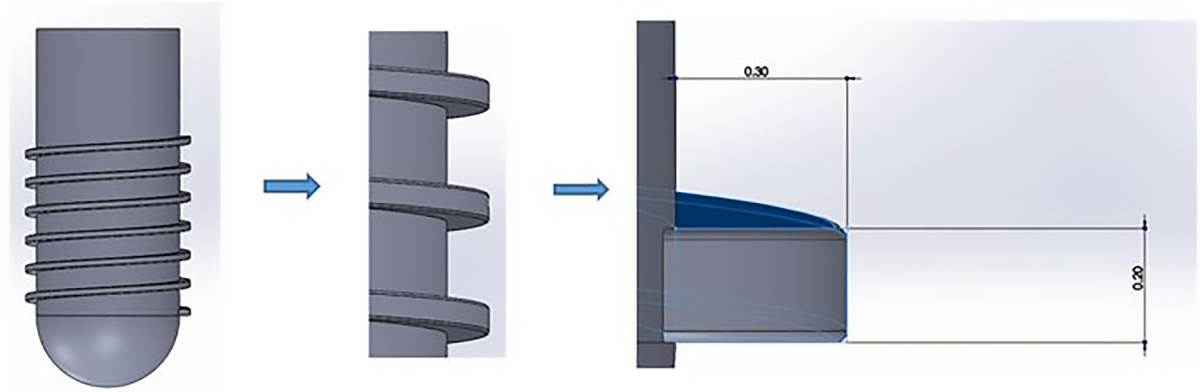


FIG. 2: 2 × 3–mm Implant thread profile

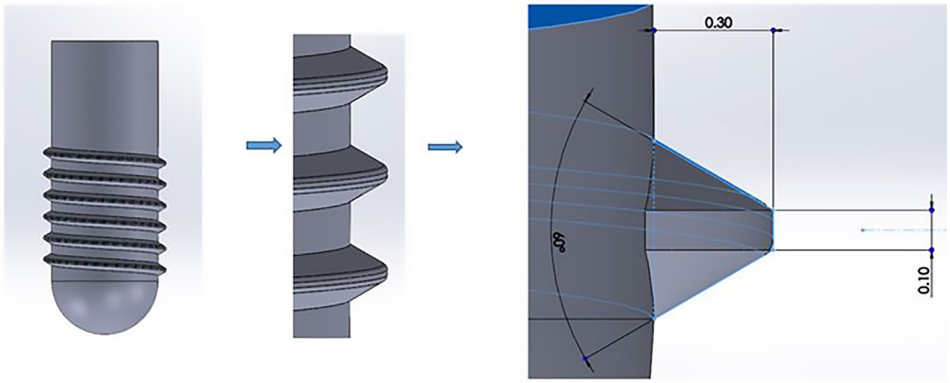


FIG. 3: 60º Trapezoidal thread profile

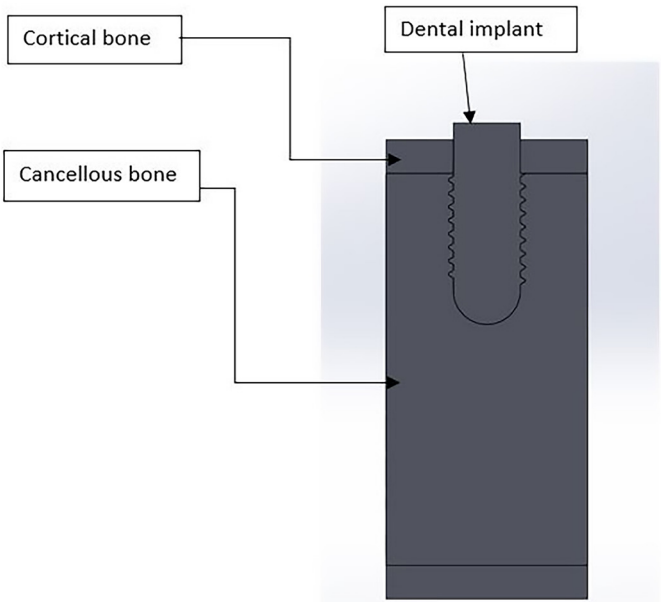


FIG. 4: Cross section of the dental implant assembly

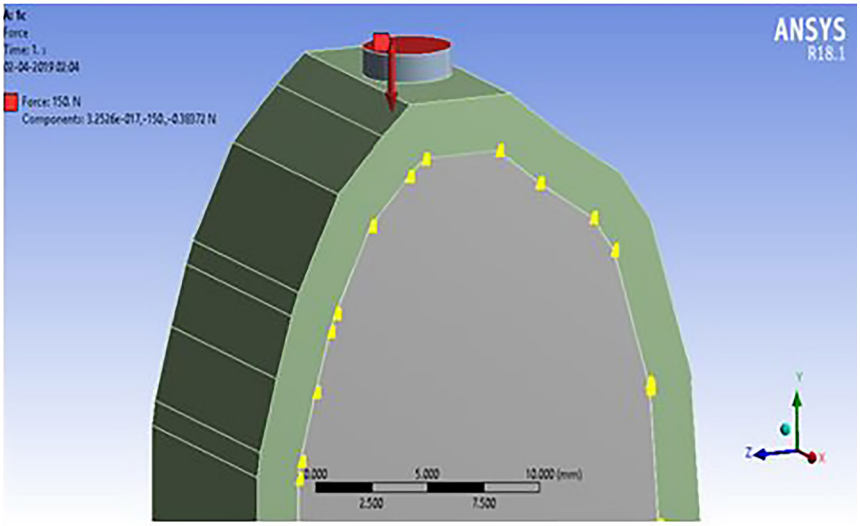


FIG. 5: Fixed boundary

The analysis performed was linear static. Partial modeling and FEAs were performed on a personal computer, with specifications of Core i7 and 16-GB random-access memory. Meshing was done using ANSYS Workbench 18.1. Another important parameter was mesh density: Results are more accurate with increased mesh density for curved geometries. We used a type of element called SOLID187, and the number of nodes and elements generated in the analysis were 591589 and 413996.

III. RESULTS AND DISCUSSION

At first, we categorized the models of dental implants based on material. The boundaries for the models were fixed, as shown in Fig. 5. We then applied a static force of 150 N (Fig. 6), corresponding to the

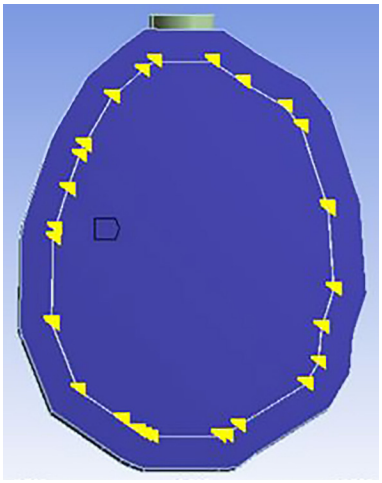


FIG. 6: Force applied

common biting force, on the tip of the screw in each simulated model.[27] After applying the load, each model was compared, on the basis of equivalent von Mises stress, to implant, cortical bone, and cancellous bone. The stainless steel was outright rejected as a possible material for dental implant because of high von Mises stress value, in the dental screw that is confirmed by the plot as shown in Fig. 7.

A negligible difference was found among von Mises stress values of commercially pure Ti and Ti6Al4V on the implant, but because stress values of commercially pure Ti were a bit less than those of Ti6Al4V, commercially pure Ti was selected to take the study forward. The value of generated stress was reduced with increasing length for small-diameter implants, shown in Fig. 8. In other words, the generated stress value was reduced in cortical bone with increasing side area for implants with small diameters. The addition of threads and varying the threads also increased the side area. The effect of thread profile on the implant was used to further categorize and narrow down models to select the best one.

Figure 9 demonstrates the effect of thread profile on the dental screw for the d4L11 implant. It can

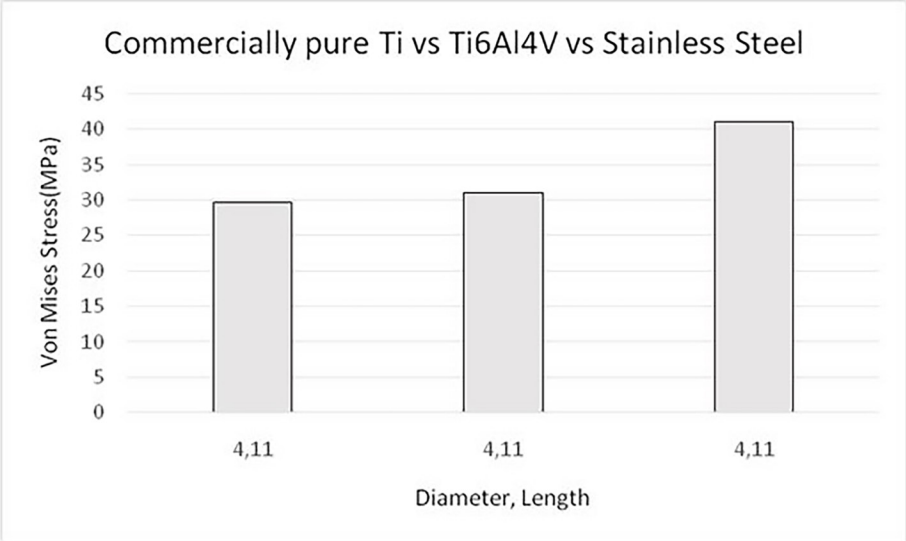


FIG. 7: von Mises stresses for d4L11 (1 × 3–mm thread)

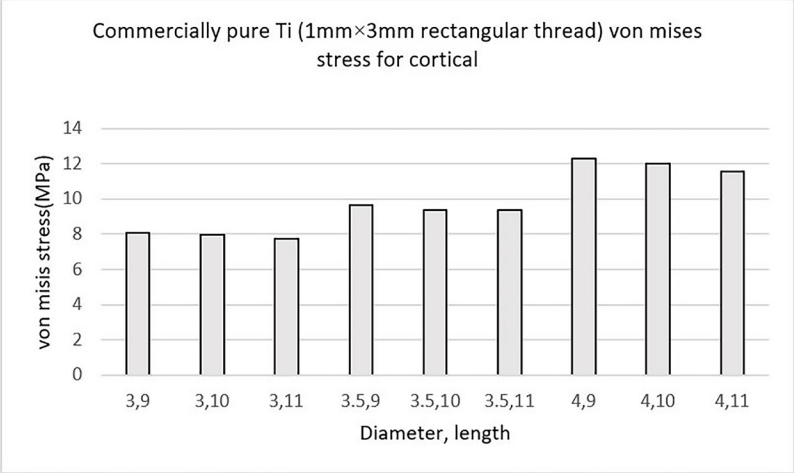


FIG. 8: von Mises stress on cortical bone for 1 × 3–mm rectangular thread

be clearly observed that a trapezoidal thread profile is best suited for dental implants. Figures 10 and 11 demonstrate the effect of thread profile on cortical and cancellous bone for the d4L11 implant. It can be clearly observed that the trapezoidal thread profile is best suited for the dental implants. The other thread profiles were rejected.

As already mentioned, a common biting force of 150 N was applied to the numerical model, and we studied stress distribution in the different dental implants. Figures 12 and 13 show the simulation of the full d4L11 implant assembly, cortical bone, and implant cross section. Implant stress can be varied with bone condition.[28] The relationship between implant length, diameter, and thread geometries was obtained from this study. This analysis may prove to be helpful in choosing implant geometry and material best used for individual patients.

IV. CONCLUSION

Within the limitations of this study, we find that commercially pure Ti can be used as dental implant material, although we did not find a big difference between commercially pure titanium and Ti6Al4V von Mises stress values. From the analysis, it was observed that implant diameter is more influential on stress value than implant length. Furthermore, we also found that a trapezoidal type of thread profile is most suitable for dental implants. The optimum solution can be predicted by considering the dynamic analysis with bony conditions.

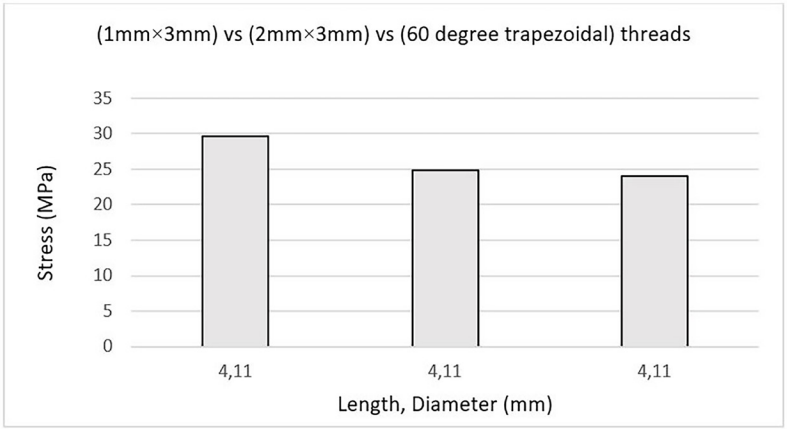


FIG. 9*:* Equivalent von Mises stress comparison on screw for different thread profiles

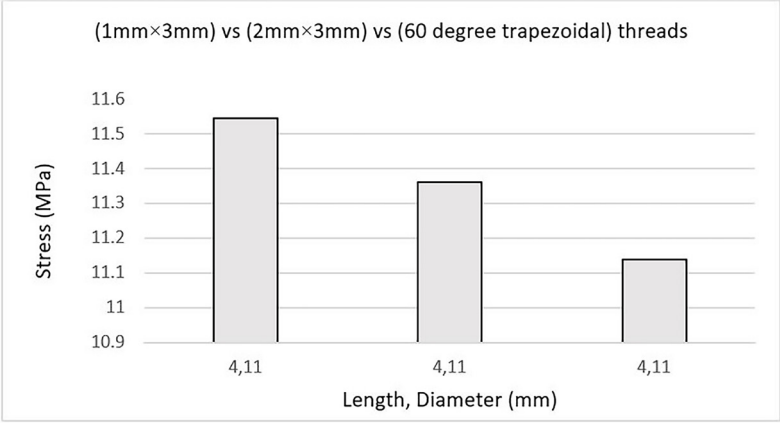


FIG. 10: Equivalent von Mises stress comparison on cortical bone for different thread profiles

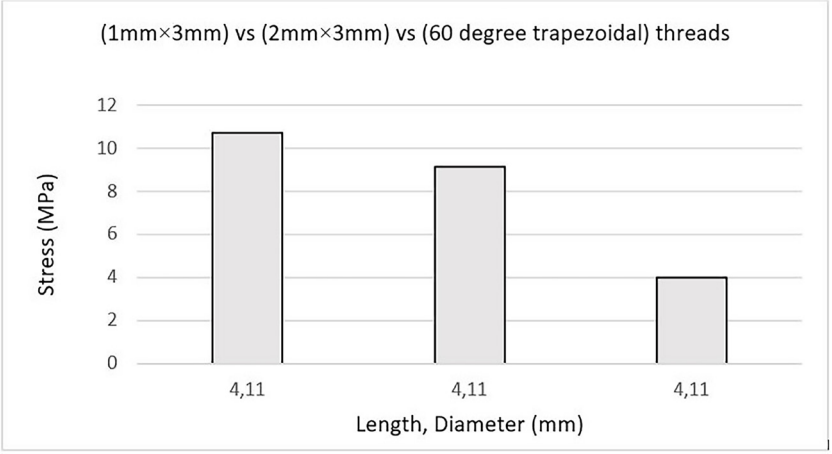


FIG. 11: Equivalent von Mises stress comparison on cancellous bone for different thread profiles

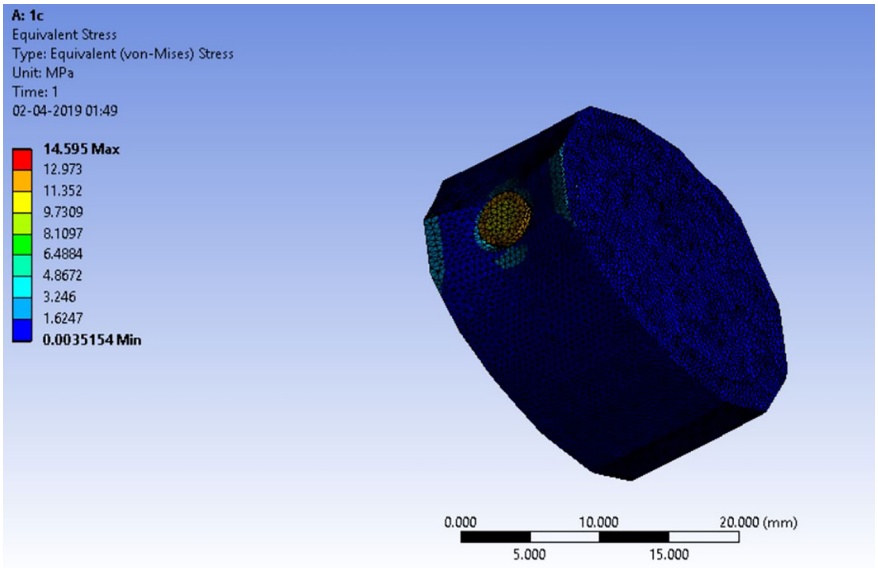


FIG. 12: Simulation of full assembly

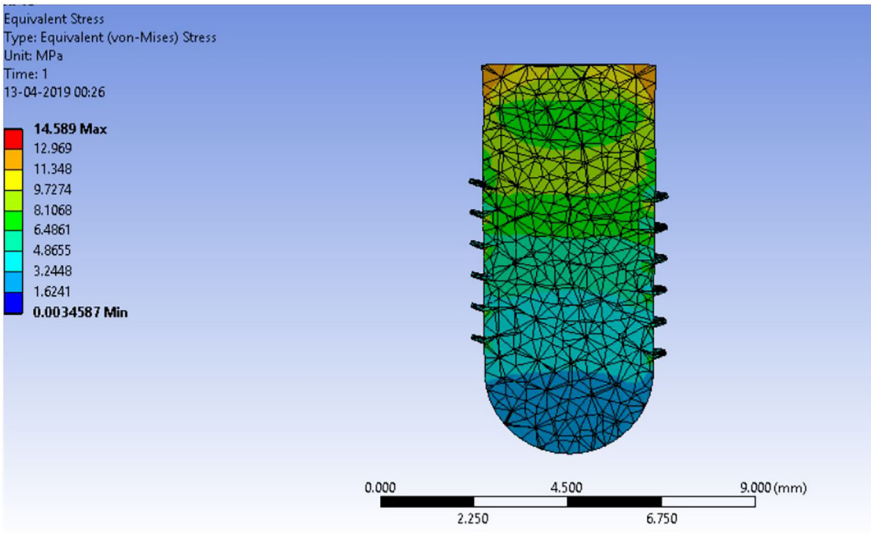


FIG. 13: Cross section of simulated dental screw

REFERENCES

1. Branemark PI. Osseointegration and its experimental background. J Prosthet Dent. 1983(50):399–410.
2. Iezzi G, Piattelli A, Mangano C, Shibli JA, Vantaggiato G, Frosecchi M, Perrotti V. Peri-implant bone tissues around retrieved human implants after time periods longer than 5 years: A retrospective histologic and histomorphometric evaluation of 8 cases. Odontology. 2014;102(1):116–21.
3. Jayme SJ, de Oliveira RR, Muglia VA, Novaes AB Jr, Ribeiro RF. The effects of different loading times on the bone response around dental implants: A histomorphometric study in dogs. Int J Oral Maxillofac Implants. 2010 May-June;25(3):473–81.
4. Javed F, Ahmed HB, Crespi R, Romanos GE. Role of primary stability for successful osseointegration of dental implants: Factors of influence and evaluation. Interv Med Appl Sci. 2013;5(4):162–7.
5. Shokri M, Daraeighadikolaei A. Measurement of primary and secondary stability of dental implants by resonance frequency analysis method in mandible. Int J Dent. 2013;2013:506968.
6. Menicucci G, Ceruti P, Barabino E, Screti A, Bignardi C, Preti G. A preliminary in vivo trial of load transfer in mandibular implant-retained overdentures anchored in 2 different ways: Allowing and counteracting free rotation. Int J Prosthodont. 2006 Nov-Dec;19(6):574–6.
7. Figueirêdo EP, Sigua-Rodriguez EA, Pimentel MJ, Oliveira Moreira AR, Nóbilo MA, de Albergaria-Barbosa JR. Photoelastic analysis of fixed partial prosthesis crown height and implant length on distribution of stress in two dental implant systems. Int J Dent. 2014;2014:206723.
8. Reingewirtz Y, Tenenbaum H. Fixed dental prosthesis on 4.2 mm length rough implants: A case series report after an average loading time of 33 months. J Oral Implants. 2015;2015. DOI: 10.1155/2015/390974.
9. Coelho PG, Suzuki M, Guimaraes MV, Marin C, Granato R, Gil JN, Miller RJ. Early bone healing around different implant bulk designs and surgical techniques: A study in dogs. Clin Implant Dent Relat Res. 2010 Sept;12(3):202–8.
10. Leonard G, Coelho P, Polyzois I, Stassen L, Claffey N. A study of the bone healing kinetics of plateau versus screw root design titanium dental implants. Clin Oral Implant Res. 2009;20(3):232–9.
11. Demiralp KÖ, Akbulut N, Kursun S, Argun D, Bagis N, Orhan K. Survival rate of short, locking taper implants with a plateau design: A 5-year retrospective study. BioMed Res Int. 2015;197451:1–8.
12. Suzuki M, Calasans-Maia MD, Marin C, Granato R, Gil JN, Granjeiro JM, Coelho PG. Effect of surface modifications on early bone healing around plateau root form implants: An experimental study in rabbits. J Oral Maxillofac Surg. 2010 Jul 1;68(7):1631–8.
13. Coelho PG, Granato R, Marin C, Bonfante EA, Janal MN, Suzuki M. Biomechanical and bone histomorphologic evaluation of four surfaces on plateau root form implants: An experimental study in dogs. Oral Surg Oral Med Oral Pathol Oral Radiol Endodontol. 2010 May 1;109(5):e39–45.
14. Coelho PG, Bonfante EA, Marin C, Granato R, Giro G, Suzuki M. A human retrieval study of plasma-sprayed hydroxyapatite-coated plateau root form implants after 2 months to 13 years in function. J Long Term Eff Med Implants. 2010;20(4):335–42.
15. Maló P, De Araújo Nobre M, Rangert B. Short implants placed one-stage in maxillae and mandibles: A retrospective clinical study with 1 to 9 years of follow-up. Clin Im-

plant Dent Relat Res. 2007 Mar;9(1):15–21.
16. Davies, JE. Understanding peri-implant endosseous healing. J Dent Educ. 2003;67(8):932–49.
17. Berglundh T, Abrahamsson I, Lang NP, Lindhe J. De novo alveolar bone formation adjacent to endosseous implants: A model study in the dog. Clin Oral Implant Res. 2003 May;14(3):251–62.
18. Coelho PG, Suzuki M, Marin C, Granato R, Gil LF, Tovar N, Jimbo R, Neiva R, Bonfante EA. Osseointegration of plateau root form implants: Unique healing pathway leading to haversian-like long-term morphology. In: Engineering mineralized and load bearing tissues. Cham, Switzerland: Springer; 2015. p. 111–28.
19. Misch CE, Bidez MW, Sharawy M. A bioengineered implant for a predetermined bone cellular response to loading forces. A literature review and case report. J Periodontol. 2001 Sep 1;72(9):1276–86.
20. Roy S, Panda D, Khutia N, Chowdhury AR. Pore geometry optimization of titanium (Ti6Al4V) alloy, for its application in the fabrication of customized hip implants. Int J Biomater. 2014. DOI:10.1155/2014/313975.
21. Roy S, Dey S, Khutia N, Chowdhury AR, Datta S. Design of patient specific dental implant using FE analysis and computational intelligence techniques. Appl Soft Comp. 2018 Apr 1;65:272–9.
22. Saidin S, Kadir MR, Sulaiman E, Kasim NH. Effects of different implant–abutment connections on micromotion and stress distribution: Prediction of microgap formation. J Dent. 2012 Jun 1;40(6):467–74.
23. Sevimay M, Turhan F, Kiliçarslan MA, Eskitascioglu G. Three-dimensional finite element analysis of the effect of different bone quality on stress distribution in an implant-supported crown. J Prosthetic Dent. 2005 Mar 1;93(3):227–34.
24. Karl M, Dickinson A, Holst S, Holst A. Biomechanical methods applied in dentistry: A comparative overview of photoelastic examinations, strain gauge measurements, finite element analysis and three-dimensional deformation analysis. Eur J Prosthodon Res Dent. 2009 Jun;17(2): 50–7.
25. Watanabe F, Hata Y, Komatsu S, Ramos TC, Fukuda H. Finite element analysis of the influence of implant inclination, loading position, and load direction on stress distribution. Odontology. 2003 Sep 1;91(1):31–6.
26. Mammadzada S, Artunç C, Sen F, Güngör M, Tekin U, Çömlekoğlu E. Effect of abutment and implant shapes on stresses in dental applications using FEM. Mathemat Comp Appl. 2011 Aug;16(2):546–55.
27. Kasani R, Attili BK, Dommeti VK, Merdji A, Biswas JK, Roy S. Stress distribution of overdenture using odd number implants—A finite element study. J Mech Beh Biomed Mater. 2019 Oct 1;98:369–82.
28. Roy S, Das M, Chakraborty P, Biswas JK, Chatterjee S, Khutia N, Saha S, Chowdhury AR. Optimal selection of dental implant for different bone conditions based on the mechanical response. Acta Bioeng Biomech. 2017;19(2):11–20.